

2019

ممثل

قسم هندسة الطاقة الكهربائية

مصطفى القريشي

اسئلة سنوات سيركت ٢

Al-Balqa' Applied University
Faculty of Engineering Technology
Electrical circuits 2

جامعة البلقاء التطبيقية
كلية الهندسة التكنولوجية

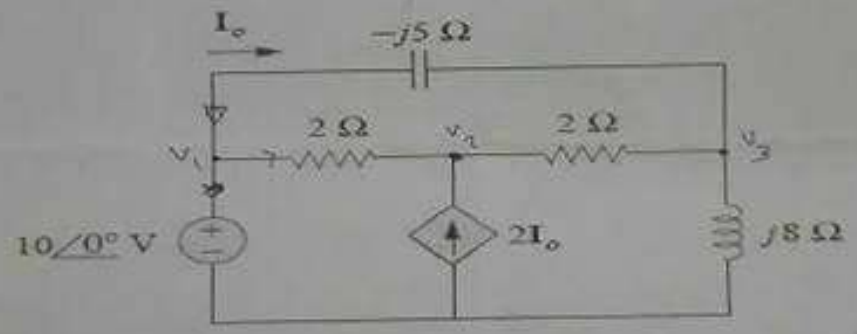
First exam 01-10-2016

Problem 1 [4 points]
For the following pair of sinusoids, answer the following:

$V_1(t) = 4 \sin(377t + 10^\circ)$ and $V_2(t) = -4 \cos(377t + 10^\circ)$

1- V_2 leads V_1 by angle of 270°
2- The voltage (v_1) value at t= 5μs is: ~~-3.93~~ V
3- The frequency in Hertz is : 60 Hz
4- The period in second is: 0.166 s

Problem 2 [8 points]
For the circuit shown in figure below :



a) Write the nodal equations for the circuit?

$\frac{V_1 - 10\angle 0^\circ}{-j5} = 0$ ... (1)

$\frac{V_1 - V_2}{2} + 2I_o = \frac{V_2 - V_3}{2}$

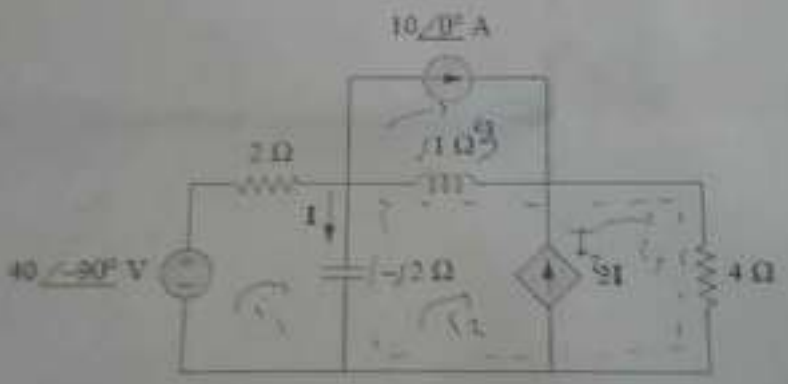
$\frac{V_2 - V_3}{2} + \frac{V_3 - V_1}{-j5} = \frac{V_3}{j8}$

b) Find the values of :

V_1= 10 V V_2= _____ V_3= _____

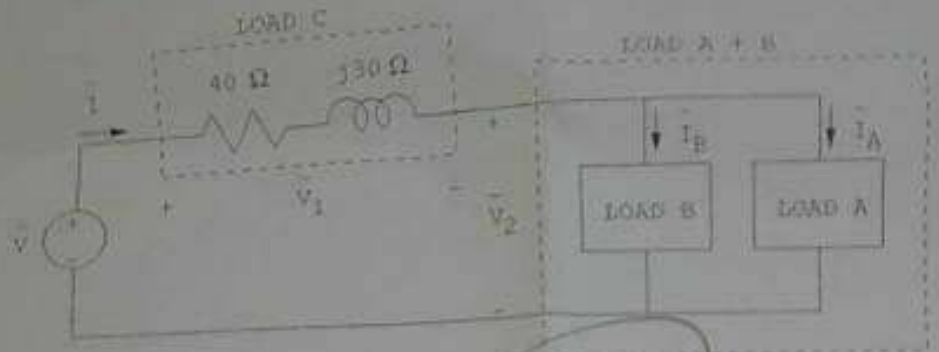
Problem 3 [8points]

For the circuit shown below.



a) Write the mesh equations for the circuit?

Q3. Fill in the missing information in the table below for each specified load and for the total load supplied by the source.



2/10

LOAD	S (VA)	P (W)	Q (VAR)	Φ (deg)	Pf (lead/lag)	I (A)	V (rms)
A	250	200	150	143.13°	0.8 (lead)	1.4 ∠-143°	250
B	300	0	300	90°	0 (lag)	1.2 ∠-90°	250
A+B	550	200	450	113.13°	0.36	2.2 ∠113.13°	250
C	∝	90	∝	∝	∝	2 ∠115°	∝
A+B+C	∝	∝	∝	∝	∝	∝	∝

Pro 1 :- $V_1(t) = 4\sin(377t + 10^\circ)$ V

$V_2(t) = -4\cos(377t + 10^\circ)$ V

$\rightarrow V_1(t) = 4\cos(377t + 10^\circ - 90^\circ)$

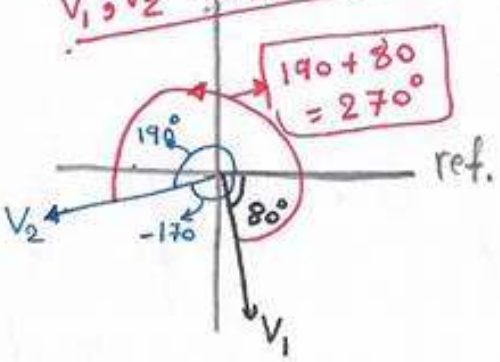
$\rightarrow V_2(t) = 4\cos(377t + 10 \pm 180^\circ)$

$\rightarrow V_1(t) = 4\cos(377t - 80^\circ)$

$\rightarrow V_2(t) = 4\cos(377t - 170^\circ)$

$\rightarrow$ or $V_2(t) = 4\cos(377t + 190^\circ)$

هنه الزاوية التي بين V_1 و V_2



① V_2 lead V_1 by **270°** ✓

② The Voltage (V_1) Value at $t = 5\,ms$ is: **3.53 Volt** ✓

$V_1(t) = 4\cos(377t - 80^\circ)$

$\rightarrow V_1(5ms) = 4\cos(377 * 5\times10^{-3}\,ms - 80^\circ)$

$= 4\cos(1.885 - 80^\circ)$ نحولها إلى degree

$= 4\cos(108 - 80) = 3.53$

$rad * \frac{180}{\pi} = deg$

$\Rightarrow 1.885 \times \frac{180}{\pi} = 108^\circ$

③ The frequency in Hertz is : **60 Hz**

$f = \frac{1}{T} = \frac{2\pi}{2\pi} = \frac{377}{2\pi} = 60\ Hz$

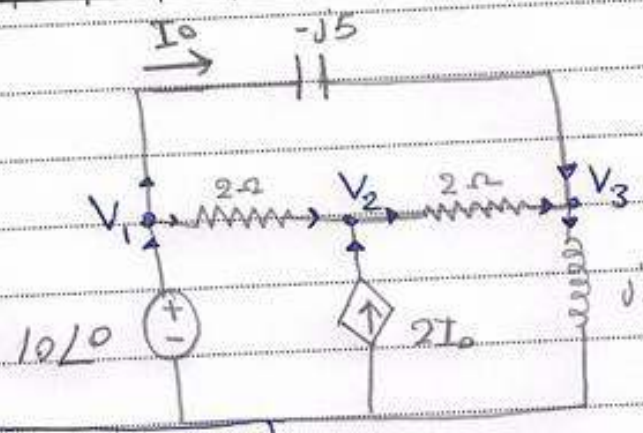
#Spark-Team #أبو-الخير-وأخواتها

④ The period in second is: **0.01667 sec**

$T = \frac{1}{f} = \frac{1}{60} = 0.01667$

Subject

Prob. 2



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#فريق سبارك

(a) $V_1 = 10\angle 0$ — (1)

$$\frac{V_1 - V_2}{2} + 2I_o = \frac{V_2 - V_3}{2} \rightarrow \frac{10\angle 0 - V_2}{2} + 2I_o = \frac{V_2 - V_3}{2} \quad (2)$$

$$\frac{V_2 - V_3}{2} + \frac{V_1 - V_3}{-j5} = \frac{V_3}{j8} \rightarrow \frac{V_2 - V_3}{2} + \frac{10\angle 0 - V_3}{-j5} = \frac{V_3}{j8} \quad (3)$$

(b) $V_1 = 10\angle 0$ — V_2 and V_3

From eq 2: $2I_o \Rightarrow I_o = \frac{V_1 - V_3}{-j5} \Rightarrow \left(\frac{10\angle 0 - V_2}{-j5}\right) \times 2 = -j4 - j0.4V_3$

$$\Rightarrow \left(\frac{10\angle 0 - V_2}{2} + j4 - j0.4V_3 = \frac{V_2 - V_3}{2}\right) * 10 \quad \text{بالضرب في 10}$$

$$\rightarrow 50 - 5V_2 + j40 - j4V_3 = 5V_2 - 5V_3$$

$$\rightarrow -10V_2 + (5 - j4)V_3 = -50 - j40 \quad (1)$$

Subject / /

From eq (3) $\left(\frac{V_2 - V_3}{2} + \frac{10\angle 0 - V_3}{-j5} = \frac{V_3}{j8}\right) * (j40)$ بالضرب في

$\rightarrow j20V_2 - j20V_3 + 80 + 8V_3 = 5V_3$

$\rightarrow j20V_2 + (3 - j20)V_3 = 80$ ——— (2)

$$\begin{bmatrix} -10 & 5-j4 \\ j20 & 3-j20 \end{bmatrix} \begin{bmatrix} V_2 \\ V_3 \end{bmatrix} = \begin{bmatrix} -50-j40 \\ 80 \end{bmatrix}$$

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فريق الشرح والتوضيح

$\Delta = (-10 * (3-j20)) - (j20 * (5-j4)) = (-110 + j100)$

$\Delta_2 = \begin{vmatrix} -50-j40 & 5-j4 \\ 80 & 3-j20 \end{vmatrix} = [(-50-j40)+(3-j20)] - [(5-j4)*80]$

$\Delta_2 = (-1350 + j1200)$

$\Delta_3 = \begin{vmatrix} -10 & -50-j40 \\ j20 & 80 \end{vmatrix} = (-10 \times 80) - ((-50-j40) * j20)$

$\Delta_3 = -1600 + j1000$

$\therefore V_2 = \frac{\Delta_2}{\Delta} = \frac{(-1350 + j1200)}{(-110 + j100)} = 12.15 + j0.136$ Volt

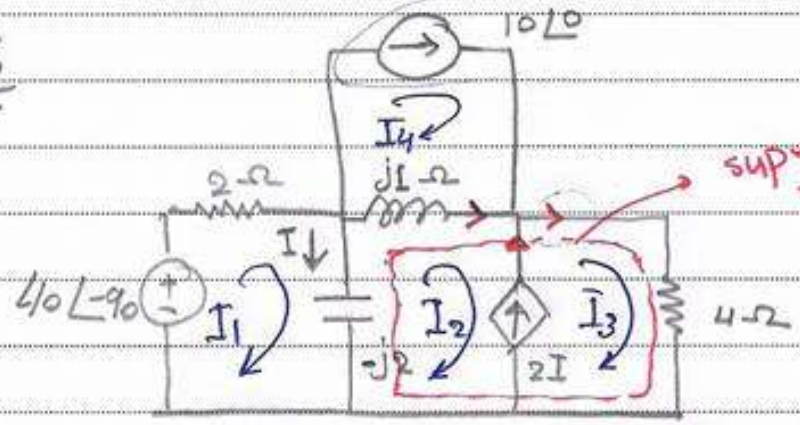
$= 12.15 \angle 0.64$ Volt

$V_3 = \frac{\Delta_3}{\Delta} = 12.49 + j2.26$ Volts

$= 12.7 \angle 10.25$ Volt

Subject ..... / /

Prob 3



(a) supermesh eq: $I_2 + 2I = I_3$; $I = I_1 - I_2$

$\therefore -I_2 + 2I_1 = I_3$ — (1)

Mesh 1 → $j40 + (2 + j2)I_1 + j2I_2 = 0$ — 2 في المعادلة

$(1 - j)I_1 + jI_2 = -j20$ — (2)

Mesh 4 → $I_4 = 10\angle 0°\ A$ — (3)

Mesh 2, 3 → $-jI_2 + j2I_1 - jI_4 + 4I_3 = 0$

$I_4 = 10\angle 0°$

$I_3 = -I_2 + 2I_1$ * (4)

$4I_3 = -4I_2 + 8I_1$

$-jI_2 + j2I_1 - j10 - 4I_2 + 8I_1 = 0$

$(8 + j2)I_1 + (-4 - j)I_2 = j10$ — (4)

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Subject

(b) $I_1, I_2, I_3, I_4 = ???$

$$I_4 = 10\angle 0 \quad -(1)$$

From eq. 2 + 4

$$\begin{bmatrix} 1-j & j \\ 8+j2 & (4-j) \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} -j20 \\ j10 \end{bmatrix}$$

$$\Delta = \begin{bmatrix} 1-j & j \\ 8+j2 & (-4-j) \end{bmatrix} = [(1-j)*(-4-j)] - [(8+j2)*j]$$

$$\boxed{\Delta = -3 - j5}$$

$$\Delta_1 \begin{vmatrix} -j20 & j \\ j10 & (4-j) \end{vmatrix} = [(-j20*(-4-j)] - (j10 \times j)$$

$$\boxed{\Delta_1 = -10 + j80}$$

$$\Delta_2 = \begin{vmatrix} 1-j & -j20 \\ 8+j2 & j10 \end{vmatrix} = ((1-j*j10) - ((-j20)*(8+j2))$$

$$\boxed{\Delta_2 = -30 + j170}$$

$$I_1 = \frac{\Delta_1}{\Delta} = \frac{-10+j80}{-3-j5} = -10.88 - j8.53\,A$$

$$\boxed{I_1 = 13.83\angle -141.9\,A}$$

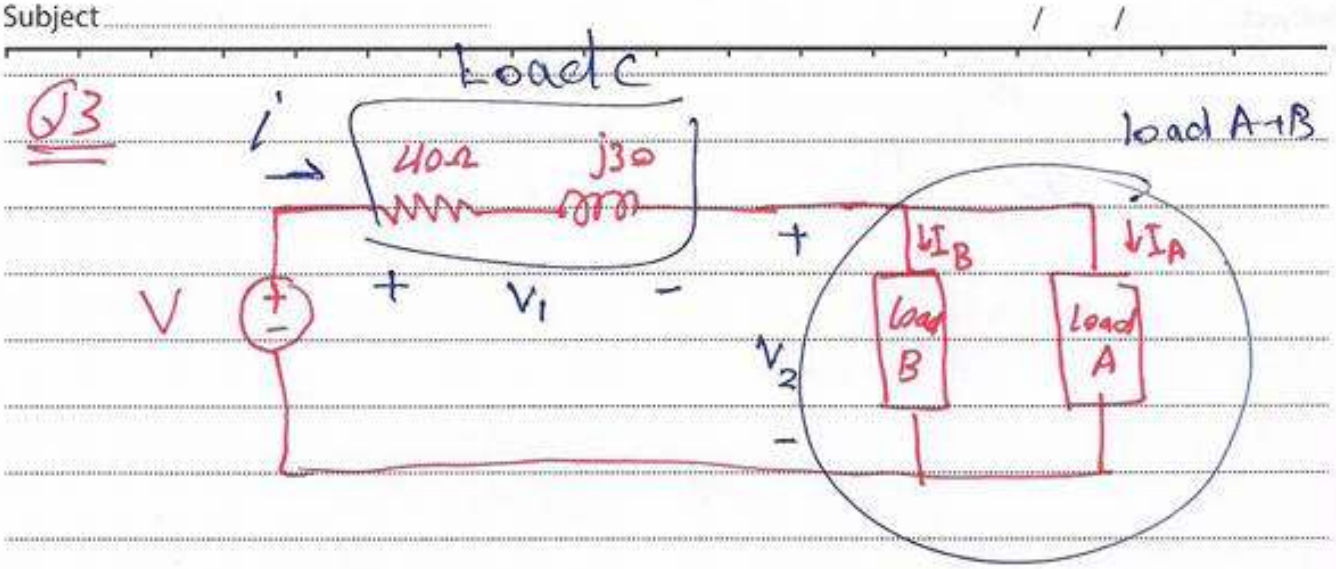
$$I_2 \frac{\Delta_2}{\Delta} = -22.35 - j19.4 = 29.6\angle -139.02\,A$$

$$\boxed{I_3 = -I_2 + 2I_1 = 2.415\angle 76.11\,A}$$

$$\boxed{I_4 = 10\angle 0}$$

Subject / /

Q3



load	S (VA	P (W)	Q (VAR	∅ lag	PF lead/lag	I(A)	Vrms
A	250	200	-j150	-36.869°	0.8 lead	1	250
B	300	zero	j300	90°	0 lag	1.2	250
A+B	250	200	j150	36.87	0.8 lag	1	250
C	50	40	j30	36.869	0.8 lag	1	50
A+B+C	300	240	j180	36.869	0.8 lag	1	300

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Note: lead ⟶ ⊖
lag ⟶ ⊕

Subject ..... / /

Load A: $S_A = 250\,VA$, $P.f = 0.8$ lead, $V_{r.m.s} = 250$

① $S = I_{r.m.s} \times V_{r.m.s} \Rightarrow I_{r.m.s} = \frac{S}{V_{r.m.s}} = \frac{250}{250} = 1A$

② $\cos(\phi) = P.f \rightarrow \phi = \cos^{-1}(P.f) = \cos^{-1}(0.8) = -36.869^\circ$

(0.8)

③ $P = S\cos(\alpha) = 250\cos(-36.869) = 200\,W$

④ $Q = jS\sin(\phi) = j250\sin(-36.869) = -j150$

Load B: $Q_B = 300$, $PF = 0$ lag , $V_{r.m.s} = 250$ ⟶ parallel

① $\cos^{-1}(0) = +90^\circ$

② $S = Q\sin(\phi) = 300\sin(90) = 300\,VA$

③ $P = S\cos(\phi) = 300\cos(90) = zero$ (zero)

④ $S = V_{r.m.s} * I_{r.m.s} \Rightarrow I_{r.m.s} = \frac{S}{V_{r.m.s}} = \frac{300}{250} = 1.2A$

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#فريق_الشرارة

Load A+B rectangular نستخرج جمعهم لأنهم P, Q ملاحظة
Polar بينما $ لأنستخرج جمعهما

① $P_{A+B} = P_A + P_B = 200 + 0 = 200$

② $Q_{A+B} = Q_A + Q_B = -j150 + j300 = j150$

③ $S = P + jQ = 200 + j150 = \sqrt{200^2 + 150^2} = 250\,VA$

Subject / /

④ $\phi = \tan^{-1}\left(\frac{Q}{P}\right) = \tan^{-1}\left(\frac{150}{200}\right) = 36.87$

⑤ P.f $= \cos(\phi) = \cos(36.87) = 0.8$ lag

⑥ $V_{r.m.s} = 250$ V → Parallel على توازي

⑦ $I_{r.m.s} = \frac{S}{V_{r.m.s}} = \frac{250}{250} = 1$ A

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load c $I_{r.m.s} = 1$, $R = 40\,\Omega$, $X_L = j30\,\Omega$

① $P = I_{r.m.s}^2 * R = 1^2 * 40 = 40$ W

② $Q = I_{r.m.s}^2 * X_L = 1^2 * j30 = j30$

③ $S = P + jQ = 40 + j30 = \sqrt{40^2 + 30^2} = 50$

④ $\phi = \tan^{-1}\left(\frac{Q}{P}\right) = \tan^{-1}\left(\frac{30}{40}\right) = 36.869^\circ$

⑤ P.F $= \cos(\phi) = \cos(36.869) = 0.8$ lag

⑥ $S = VI \Rightarrow V = \frac{S}{I} = \frac{50}{1} = 50$ V

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(9)

Load A+B+C

① $P_T = 200 + 0 + 40 = 240\ W$

② $Q_T = -j150 + j300 + j30 = j180$

③ $S_T = \sqrt{240^2 + 180^2} = 300\ VA$

④ $\theta = \tan^{-1}\left(\frac{Q}{P}\right) = \tan^{-1}\left(\frac{180}{240}\right) = +36.86^\circ$

⑤ $P.F = \cos(\theta) = \cos(36.86) = 0.8$ Lag

⑥ $V_{T\ r.m.s} = V_{A+B} + V_C = 250 + 50$ (توالي)

$= 300\ V$ (الجهد على التوالي لازم يجمع)

⑦ $S = V_{r.m.s}\ I_{r.m.s} \Rightarrow I_{r.m.s} = \frac{S}{V_{r.m.s}}$

$\Rightarrow I_{r.m.s} = \frac{300}{300} = 1\ A$

Problem 1.[8 points]

For the following pair of sinusoids, answer the following:

$V_1(t) = 4 \sin(377t + 10^\circ)$ and $V_2(t) = -4 \cos(377t + 10^\circ)$

1- V_2 leads V_1 by angle of 116°

2- The voltage (v_1) value at t= 5μs is: 3,99

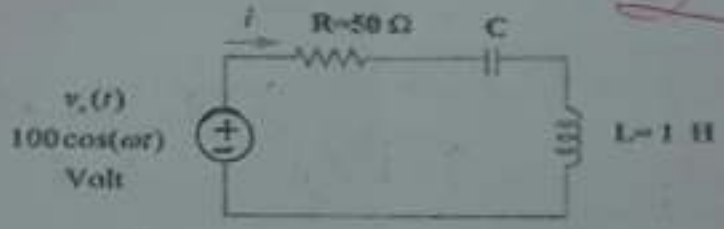
3- The frequency in Hertz is : 60,0011z

4- The period in second is: 1,0167 s

Consider the circuit below which is operating with a sinusoidal source.

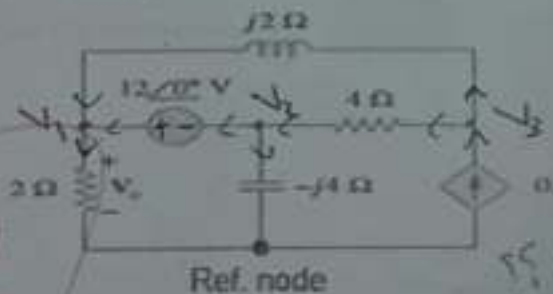
5. The value of the capacitance C, if $i(t) = \sqrt{2} \cos(\omega t - 45^\circ)$ A and ω= 100 rad/sec -5 mF

6. The value of ω, if $i(t) = \sqrt{2} \cos(\omega t + 45^\circ)$ A and C= (5/3)mF 25



Problem 2[8 points]

For the circuit shown in figure below :



a) Write the nodal equations for the circuit?

$12\angle 0 = V_1 - V_2$

$(0.5 + .5j)V_1 + (.25 + .25j)V_2 + (-.25 - .25j)V_3 = 0$

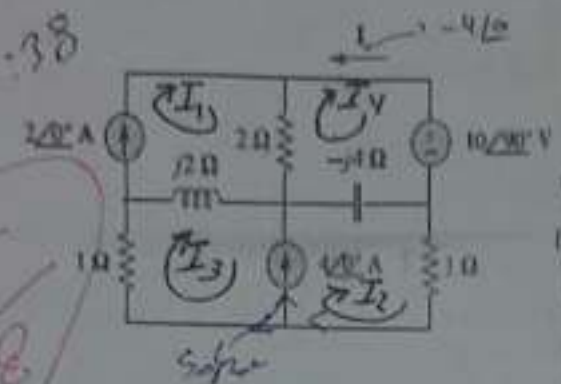
$(-.5 - .5j)V_1 + (-.25)V_2 + (.25 - .5j)V_3 = 0$

b) Find the values of:

$V_1 = 11.615$ $V_2 = -2.35$ $V_3 = -7.58 - 4j$

Problem 3 [8 points]

For the circuit shown in figure below :



a) Write the mesh equations for the circuit?

$I_2 = 4\angle 0 + I_3$

$(2+2j)I_1 + 2I_4 - 2jI_3 = 0$

$-2I_1 - 4jI_2 + (2-4j)I_4 = 10\angle 90$

$-2jI_1 + (1-4j)I_2 + (1-2j)I_3 - 4jI_4 = 0$

b) Find the values of:

$I_1 = 2\angle 0°$ A $I_2 = (6-6j)$ A $I_3 = (2-6j)$ A

$I_4 = -4\angle 0° = (4\angle 0°)$ A

Problem 4 [10 points]

For the circuit shown in figure below

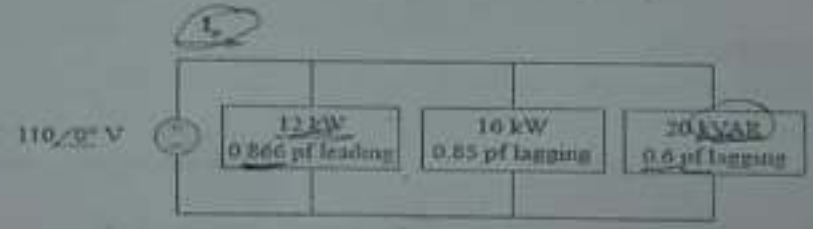
The total active power is ---23.83 W---

The total reactive power is ---0---

The total complex power is ---$S = P + jQ$ = 23.8---

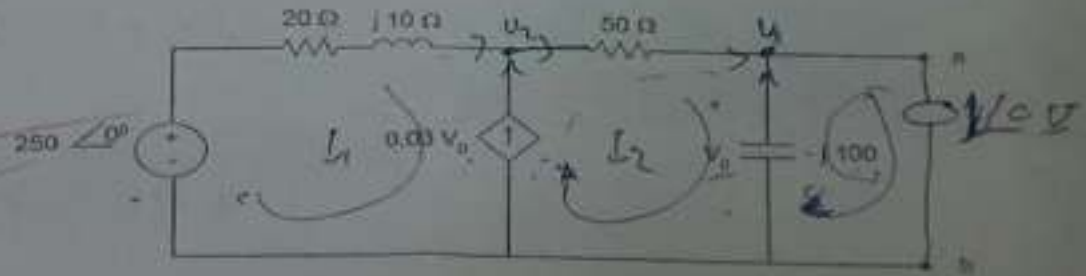
The power factor at the source is ---1---

The value of I_0 is ---(4.334) A---



Problem 5 [6 points]

Find the Thevenin's equivalent between a and b



Q1 (A)

$V_1(t) = 4 \sin(377t + 10^\circ)$

$V_2(t) = -4 \cos(377t + 10^\circ)$

Ahmed Al Khaqani

① V_2 Lead V_1 by 270°

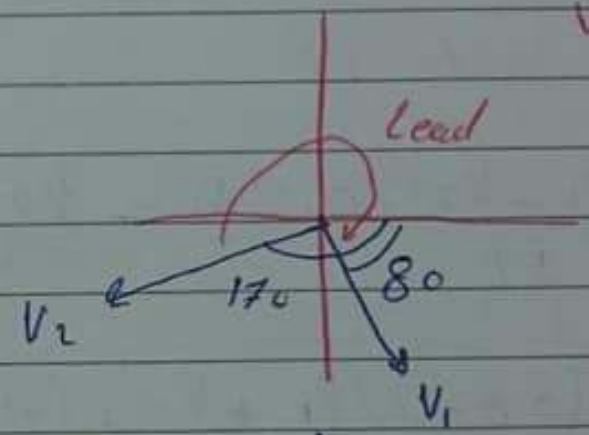
$V_1(t) = 4 \cos(377t - 80^\circ)$

$V_2(t) = 4 \cos(377t - 170^\circ)$

Lead

$170 - 80 = 90$

$360 - 90 = 270$



② $V_1(SHS) = 4 \sin(377(SH) + 10^\circ)$

نحول من راد إلى درجة

$deg = \frac{180}{\pi} * rad$

$deg = \frac{180}{\pi} * 377 * SH = 0.108$

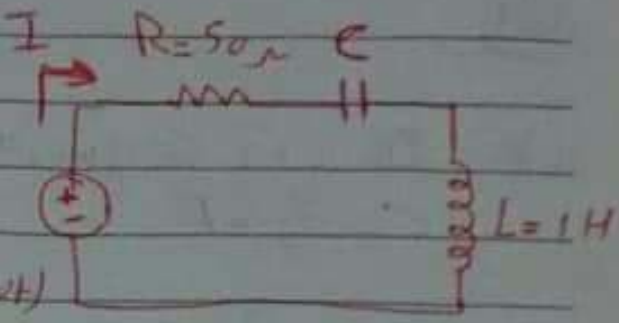
$V_1 = 0.7$ Volt

③ $f = \frac{w}{2\pi} = \frac{377}{2\pi} = 60$ Hz

④ $T = \frac{1}{f} = \frac{1}{60} = 0.0166$ s

B

Q1



(a) $i(t) = \sqrt{2}\cos(wt - 45)$

$w = 100$ rad/sec

Find C ??

Sol:- $V(t) = 100\angle 0$ $\quad I(t) = \sqrt{2}\angle -45$

$X_L = jwL = j(100)1 = (j100) \rightarrow X_L$

$V(t) = I(t)\,Z \rightarrow Z = \frac{V(t)}{I(t)} = \frac{100\angle 0}{\sqrt{2}\angle -45}$

$\rightarrow (Z = 50 + j50)$

$Z = R + X_L + X_C$

$50 + j50 = 50 + j100 + X_C \rightarrow (X_C = -j50)$

$X_C = \frac{1}{jwC} \rightarrow -j50 = \frac{1}{j(100)C}$

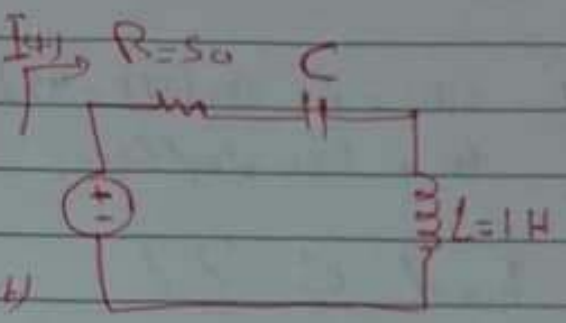
$\rightarrow (C = 2 \times 10^{-4}\ F)$

Ahmad AL-Khawaja

Q1

(b) $I(t) = \sqrt{2}\cos(wt + 45)$

$C = \frac{5}{3}\,mF$



Find w ?

Sol:-

$X_L = jwL = (jw)$ $\quad X_C = \frac{1}{jwC} = \left(\frac{-j600}{w}\right)$

$V(t) = I(t)\,Z \rightarrow Z = \frac{V(t)}{I(t)} = \frac{10\angle 0}{\sqrt{2}\angle 45}$

$Z = 50 - j50$

$Z = R + X_L + X_C \rightarrow 50 - j50 = 50 + jw - \frac{j600}{w}$

$\rightarrow jw - \frac{j600}{w} + j50 = 0$

$\rightarrow \left[w - \frac{600}{w} + 50 = 0\right] w$ Ahmed Al-Khaneg

$\rightarrow w^2 + 50w - 600 = 0$

$(w + 60)(w - 10) = 0$

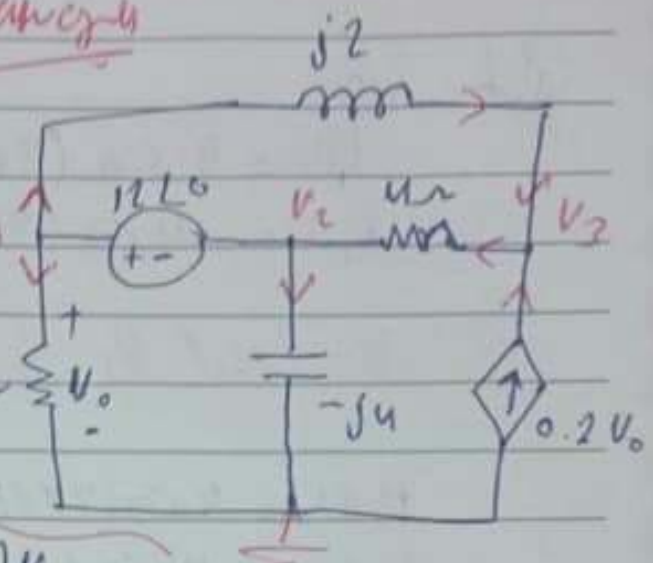
$w = -60 \qquad w = 10$

$\times$

$w = 10$ rad/sec

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Q2



at node ① + ②

$$4\left[\frac{V_3 - V_2}{4} = \frac{V_1}{2} + \frac{V_2}{-j4} + \frac{V_1 - V_3}{2j}\right]$$

$$(2 - j2)V_1 + (1+j)V_2 + (-1+j2)V_3 = 0 \quad \rightarrow ①$$

at node ③

$$V_o = V_1$$

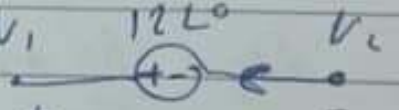
$$\left[\frac{V_1 - V_3}{j2} + 0.2V_1 = \frac{V_3 - V_2}{4}\right]4$$

$$(0.8 - j2)V_1 + V_2 + (-1+j2)V_3 = 0 \quad — ②$$

at super node

$$-V_2 - 12 + V_1 = 0$$

$$V_1 - V_2 = 12 \quad — ③$$



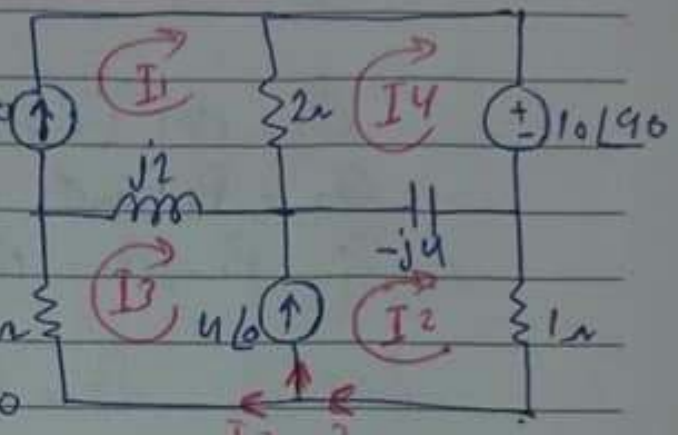
$$\begin{bmatrix} 2-j2 & 1+j & -1+j2 \\ 0.8-j2 & 1 & -1+j2 \\ 1 & -1 & 0 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 12 \end{bmatrix}$$

$V_1 = 7.68 \angle 50.19$ $\quad V_2 = 9.21 \angle 140.19$

$V_3 = 3.88 \angle 68.62$

Ahmad Al-Khawaja

Q3



at mesh (1)

$I_1 = 2\angle 0^\circ$ A

at mesh (4)

$2(I_4 - I_1) + 10\angle 90 + (-j4)(I_4 - I_2) = 0$

عوض قيمة (I_1) في المعادلة

$\longrightarrow$ $j4 I_2 + (2 - j4) I_4 = 4 - j10$ — (1)

at mesh (2) + (3)

$I_3 + (j2)(I_3 - I_1) + (-j4)(I_2 - I_4) + I_2 = 0$

عوض قيمة (I_1) في المعادلة

$(1 - j4) I_2 + (1 + j2) I_3 + (j4) I_4 = j4$ — (2)

at super mesh

$I_2 = 4\angle 0^\circ + I_3 \longrightarrow I_2 - I_3 = 4\angle 0^\circ$ — (3)

$$\begin{bmatrix} j4 & -1 & 2 + j4 \\ 1 - j4 & 1 + j2 & j4 \\ 1 & -1 & 0 \end{bmatrix} \begin{bmatrix} I_2 \\ I_3 \\ I_4 \end{bmatrix} = \begin{bmatrix} 4 - j10 \\ j4 \\ 4\angle 0 \end{bmatrix}$$

$I_1 = 2\angle 0^\circ$ A

$I_2 = 1.05\angle 18.43$ A

$I_3 = 3.018\angle 173.65$ A

$I_4 = 3.34\angle -5.71$ A

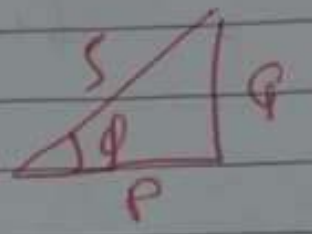
Q4

Load 1 $P_1 = 17\,KW$ 0.866 Lead

$\phi_1 = \cos^{-1}(0.866) = 30^\circ$

$Q_1 = P_1 \tan\phi_1 = 6.928\,KVAR$

$\tilde{S}_1 = P_1 - jQ_1 = 17K - j\,6.928K\ VA$



Load 2 $P_2 = 16\,KW$ 0.85 Lag

$\phi_2 = \cos^{-1}(0.85) = 31.78^\circ$

$Q_2 = P_2 \tan\phi_2 = 9.915\,KVAR$

$\tilde{S}_2 = P_2 + jQ_2 = 16K + j\,9.915\,K\ VA$

Load 3 $Q_3 = 20\,KVAR$ 0.6 Lag

$\phi_3 = \cos^{-1}(0.6) = 53.13^\circ$

$P_3 = \frac{Q_3}{\tan\phi_3} = 15\,KW$

$\tilde{S}_3 = P_3 + jQ_3 = 15K + j\,20K\ VA$

(3) $\tilde{S}_{tot} = \tilde{S}_1 + \tilde{S}_2 + \tilde{S}_3 = 47K + j\,22.987K\ VA$

(1) $P_{tot} = 47\,KW$ (2) $Q_{tot} = 22.987\,KVAR$

$S_{tot} = 48.758 \angle 28.128$

(4) $PF = \cos(28.128) = 0.88$

Q4

(5) I.P.T

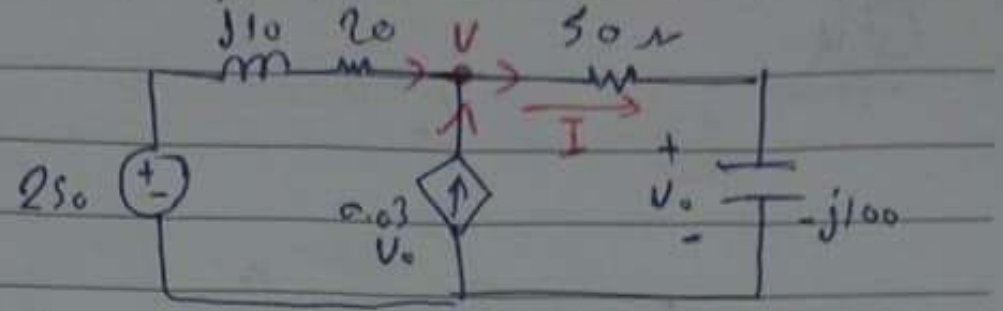
$$\tilde{S} = \tilde{V}_{rms}\,\tilde{I}_{rms}^{*} = \frac{1}{2}\tilde{V}\,\tilde{I}^{*}$$

$$\tilde{I} = \left(\frac{2\tilde{S}}{\tilde{V}}\right)^{*} = (886.52\ \angle 28.13)^{*}$$

$$\tilde{I} = 886.52\ \angle -28.13\ A$$

Ahmad Al-Khawaja

Q5



$V_o = V_{1h}$

$V_o = I(-j100)$

$V_o = \frac{V}{50-j100} \times (-j100)$

at node (V)

$$\frac{250\angle 0^\circ - V}{20+j10} + 0.03\left(\frac{V}{50-j100} \times (-j100)\right) = \frac{V}{50-j100}$$

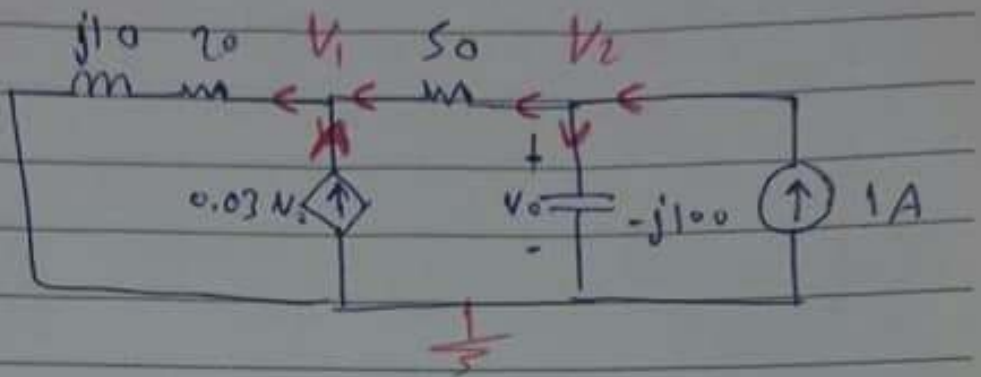
بتبسيط المعادلة

$$-0.02V = -10+j5 \longrightarrow V = 500 - j250$$

$$V_o = V_{1h} = \frac{V}{50-j100} \times (-j100) = 300 - j400 \text{ Volt}$$

Ahmed Al-Khwajah

Q5 B



at node ①

$V_x = V_2$

$$\frac{V_2 - V_1}{50} + 0.03V_2 = \frac{V_1}{20+j10}$$

بتبسيط المعادلة

$$(-0.06 + j0.02)V_1 + (0.05)V_2 = 0 \quad \text{——} ①$$

at node ②

$$\frac{V_2 - V_1}{50} + \frac{V_2}{-j100} = 1 \longrightarrow (-0.02)V_1 + (0.02 + j0.01)V_2 = 1 \quad ②$$

$$\begin{bmatrix} -0.06+j0.02 & 0.05 \\ -0.02 & 0.02+j0.01 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$V_1 = 100 - j50$ $\quad$ $V_2 = 100 - j100$

الجهد عند مصدر التيار يساوي الجهد عند الكواسع ($-j100$)

حل المقاومة

$$Z_{th} = \frac{V_2}{I_s} = \frac{100 - j100}{1} = 100 - j100$$